Ch 5.1.1-3: Leave one out and k-Fold Cross-Validation

Lecture 13 - CMSE 381

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Mon, Feb 16, 2026

Announcements

Last time:

- Exam

Announcements:

- Exam 1 grades ....hopefully soon

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Mon 2/9	
	M	2/9	***Project Day & Review***			
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16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2	HW #3 Due Sun 3/1	
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19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6

Covered in this lecture

- Validation Set
- LOO CV
- K-fold CV

- Previous versions of the course with more time I talked about the special case with linear regression. Due to time constraints, the slides below are commented out:
- Outliers
- Leverage statistic

Section 1

Validation set

What's the problem?

- How well is my ML method doing? *Model Assessment*
- Which method is best for our data?
- How many features should I use? Which ones? *Model selection*
- What is the uncertainty in the learned parameters?

First, Cross Validation. Then Bootstrap.

Training Error vs Testing Error

Training Error

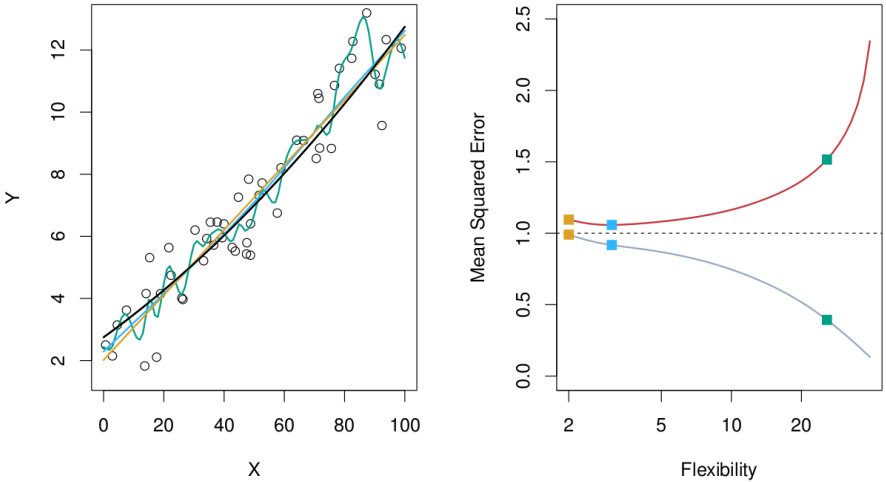
Easily calculated by applying stats learning method to the observations used in its training

- Massive under estimate
- overfitting!

Testing Error

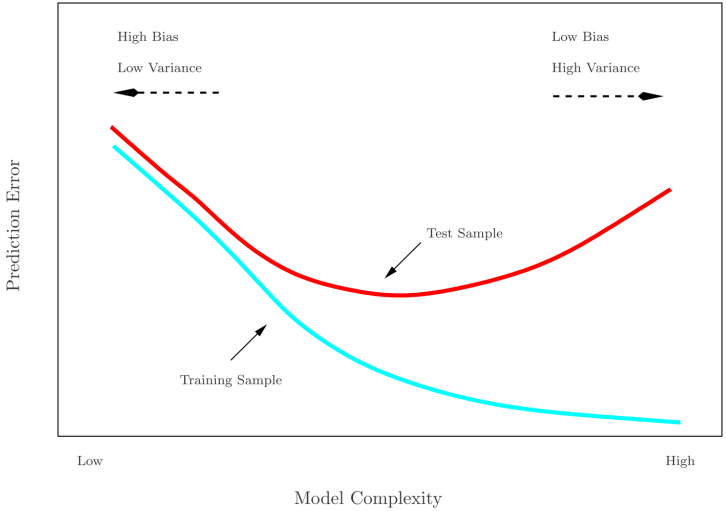
Avg error that results from using a statistical learning method to predict the response on a new observation, one that was not used in training the method.

Throw-back

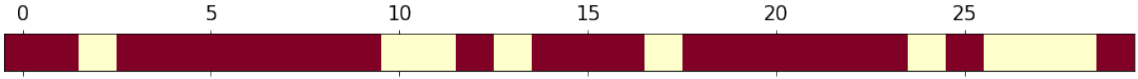


Grey is training error, red is testing error

Model tradeoffs



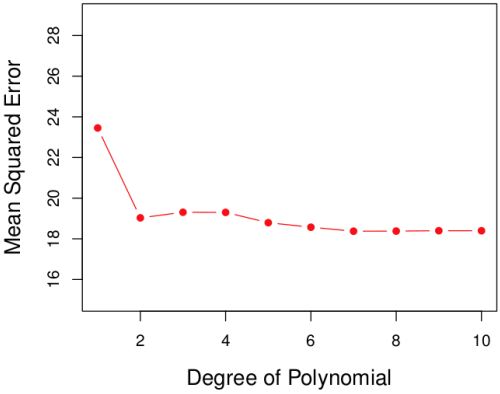
Validation set approach



- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

- This example has 30 data points
- Keep out 30% = 9 for testing in yellow
- We've done this in previous jupyter notebooks

Example with the `auto` data

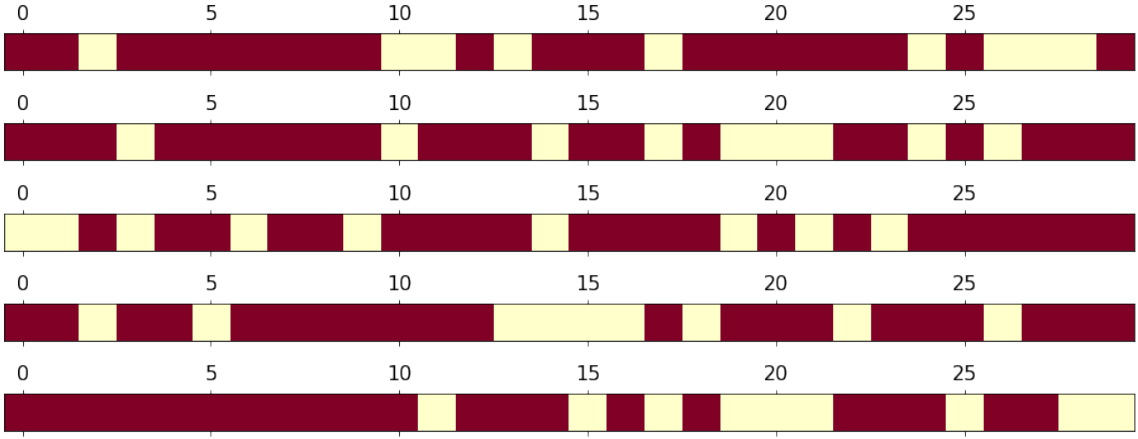


Predicting `mpg` using `horsepower`:

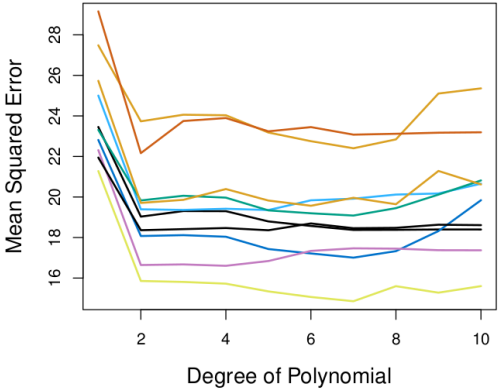
$$\texttt{mpg} = \beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2 + \cdots + \beta_p \texttt{hp}^p$$

- Shows the TEST error
- Can conclude that linear isn't enough to model the data

Rinse and repeat



Again example with auto data



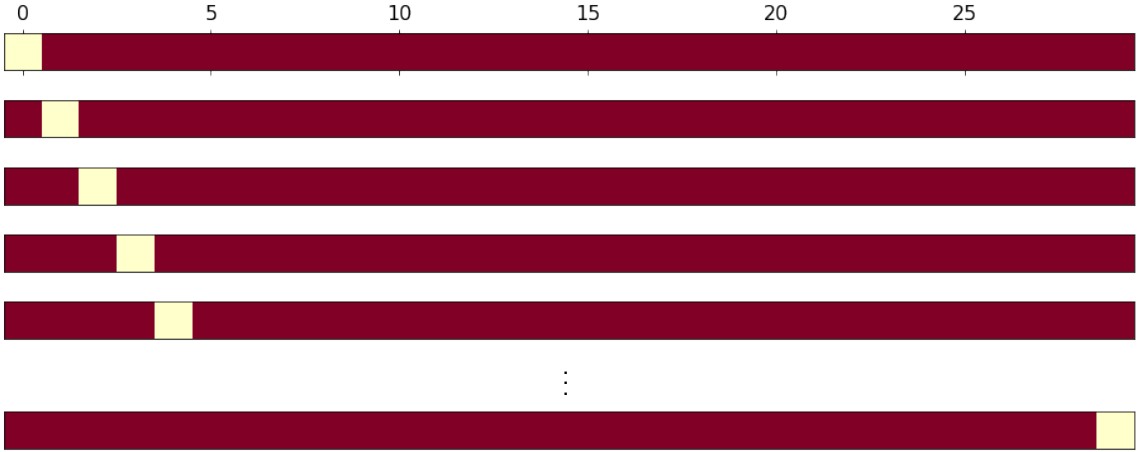
- Repeat the procedure with different splits
- Highly variable results, no concensus about the error
- Further complicated: Stats methods do worse when trained on fewer observations, so validation set error rate tends to overestimate test error rate when fitted on the whole data set.

Coding example in jupyter notebook

Section 2

Leave-One-Out Cross-Validation (LOOCV)

The idea



The idea in mathy words

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

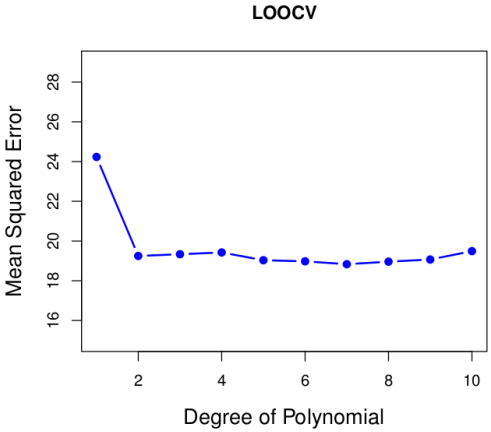
- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

- Rinse and repeat

Return the score:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i$$

Again example with `auto` data



- Point out that there is no randomness. Same every time.

Do the LOOCV coding section

LOOCV Pros and Cons

Advantages:

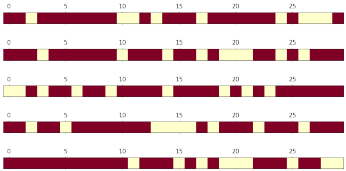
- Using almost all data every time
- Stable results: no randomness so always the same answer on a fixed data set
- Evaluation is performed with more testing data

Disadvantages:

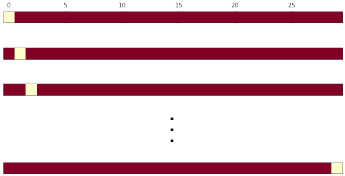
- Computationally expensive; fit the model n times
- Doesn't shake up the data enough: estimates from each fold are highly (positively) correlated so their averages have high variance.

TL;DR

Validation set



LOO-CV



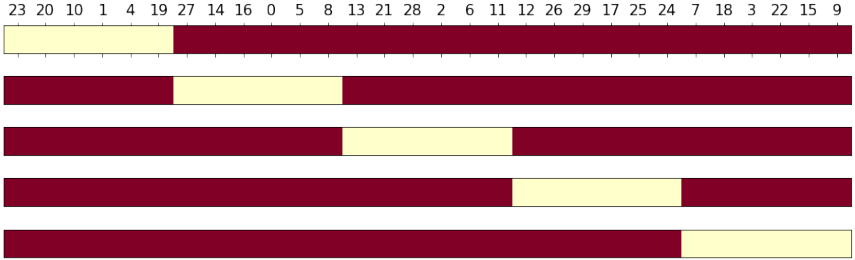
LOO-CV Score

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Section 3

k-Fold CV

The idea



- $K = 5$ in this example
- Randomly sort, split up into K subsets
- If $K = n$, then we get LOOCV

Mathy version

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

By hand first!

There are 10 students in the class, and we have data points for each. They have already been randomly permuted below. Write down the training/testing sets for a 3-fold CV

- Damien
- Alice
- Greta
- Jasmin
- Benji
- Inigo
- Frank
- Carina
- Enrique
- Hubert

Fold 1

- Test: DAGJ
- Train: BIFCEH

Fold 2

- Test:BIF
- Train: DAGJCEH

Fold 3

- Test: CEH
- Train: DAGJBIF

Coding - Building k-fold CV

Pros and Cons

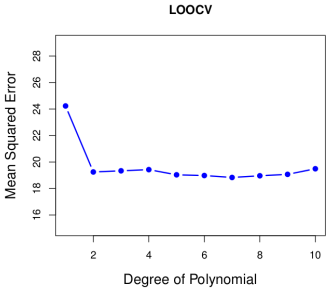
Pros:

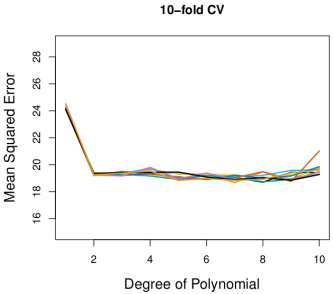
- Cheaper computation with $k = 5$ or 10 instead of n

Cons:

- Unlike LOOCV, there is randomness, but it has lower variability than validation

Comparison


LOOCV
Mean Squared Error
16
18
20
22
24
26
28
2
4
6
8
10
Degree of Polynomial


10–fold CV
Mean Squared Error
16
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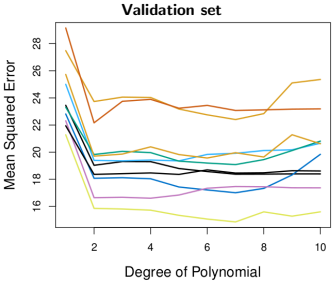
22
24
26
28
2
4
6
8
10
Degree of Polynomial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Mean Squared Error
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Degree of Polynomial

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